

Kanishka Roy

Data Analyst — SQL, Python, dbt & Experimentation

Canada — +1 403 918 8909 — roykanishka71@gmail.com

linkedin.com/in/kanishka-roy-64815319b — github.com/kroyyc — Portfolio

Profile

Data analyst and engineer with experience transforming complex datasets into reliable models, experiments, reporting, and actionable insights using SQL, Python, R, dbt, Snowflake, Power BI, and Tableau. Strong quantitative background in statistical experimentation, predictive modelling, forecasting, data modelling, and performance analysis, with production experience building and validating data pipelines and analytical workflows. Experienced partnering with technical and non-technical stakeholders to investigate business and product problems, communicate findings, and translate requirements into data-driven solutions. Currently completing an MSc in High Performance Computing with Data Science following a BSc in Applied Mathematics & Computer Science.

Technical Skills

Analytics & Experimentation: SQL, Python, R, NumPy, pandas, Statistical Analysis, Experimental Design, Hypothesis Testing, Predictive Modelling, Forecasting, Regression, Model Evaluation, A/B Experiment Analysis, Exploratory Data Analysis

Data Modelling & Engineering: dbt, Snowflake, Snowpark, Snowpipe, Streams, Tasks, ETL/ELT, Data Modelling, Data Pipelines, Data Validation, Data Quality, PostgreSQL, SQL Server, Oracle, MySQL, NoSQL

BI & Decision Support: Power BI, Tableau, Data Visualization, Dashboarding, Analytical Reporting, KPI Analysis, Business Analytics, Stakeholder Reporting, Microsoft Copilot

Data Science & Platforms: scikit-learn, TensorFlow, PyTorch, Machine Learning, Feature Engineering, Ensemble Modelling, Azure, Databricks, AWS, Git, CI/CD, Automated Testing

Professional & Engineering Experience

Symend

2023–2024

Software Engineer – Platform Dataflow Team

- Developed high-volume production data pipelines using Python and SQL to ingest, transform, validate, and structure enterprise data for downstream applications, analytics, and reporting.
- Used dbt and Snowflake to develop maintainable SQL-based transformations and analytical data models, working with Snowpark, Snowpipe, Streams, and Tasks across production data workflows.
- Worked with Power BI and Tableau to make processed data accessible for reporting and analytical use cases, supporting interpretation of business and operational performance.
- Queried and analysed relational datasets across PostgreSQL, SQL Server, Oracle, and other data platforms, investigating data quality, pipeline behaviour, anomalies, and downstream impacts.
- Partnered with engineering, product, DevOps, and client-facing stakeholders to understand requirements, investigate production problems, and translate technical findings into actionable solutions.

EquiRisk AI

2025–2026

Software Engineer – Data & AI

- Developed Python analytical workflows to ingest, transform, validate, and analyse structured and unstructured enterprise data, producing automated reporting and decision-support outputs.
- Investigated datasets and application behaviour to identify patterns, data-quality issues, and failure modes, combining quantitative analysis with deterministic validation and business rules.
- Built AI-assisted analytical workflows using LLMs, embeddings, semantic retrieval, and RAG to surface information and insights from complex data while maintaining provenance and source attribution.
- Worked with stakeholders to turn ambiguous analytical and product requirements into measurable data workflows, reporting functionality, and maintainable software solutions.

AMD Canada

2024–2025

Software Engineer (Part-Time)

- Used Python, C++, SQL, and quantitative analysis to investigate performance behaviour across CPU/GPU computing environments and structure technical data for reproducible analysis.
- Designed controlled experiments and benchmarks to compare system behaviour, identify regressions and anomalies, and quantify runtime and resource-performance trade-offs.
- Used dbt and SQL-based workflows to transform and validate technical data used in engineering analysis, alongside automated testing and reproducible analytical processes.

Formula Student – University of Edinburgh

2025–2026

Software Engineering Manager

- Developed Python-based telemetry analytics workflows that transformed high-frequency vehicle sensor measure-

ments into interpretable data for performance analysis and engineering decisions.

- Applied filtering, visualization, statistical analysis, and comparative experimentation to investigate vehicle and subsystem behaviour while communicating findings across engineering disciplines.

CalgaryToSpace – University of Calgary

2024–2026

Software Engineering Manager

- Developed telemetry and engineering-data workflows spanning sensor acquisition, validation, processing, communications, and downstream analysis for a satellite platform that was ultimately deployed.
- Coordinated multidisciplinary data and software integration, translating engineering requirements into processing and analytical workflows used to evaluate system behaviour.

Education

University of Edinburgh – School of Informatics / EPCC

2025–2026

MSc Informatics – High Performance Computing with Data Science

Graduate study spanning high-performance data analytics, data management, statistical and computational analysis, parallel computing, and scientific software development. Expected graduation: November 2026.

University of Calgary

2019–2025

BSc Applied Mathematics & Computer Science

Interdisciplinary quantitative training in statistics, probability, data science, machine learning, databases, numerical methods, optimization, mathematical modelling, algorithms, and software engineering.

Selected Analytics & Research

MSc Dissertation – University of Edinburgh

2026

Investigating the Numerical Stability of Polynomial Basis Representations in Constrained Probabilistic Inference

- Built a reproducible Python/NumPy/pandas experimentation framework across approximately 600 benchmark records to compare accuracy, runtime, conditioning, sensitivity, and constraint preservation across alternative computational approaches.
- Designed controlled experiments and independent validation methods to identify failure modes, analyse performance differences, and distinguish meaningful effects from numerical or implementation artifacts.
- Worked closely with University of Edinburgh faculty and PhD researchers on experimental design, quantitative analysis, and interpretation of results; research is being prepared for publication.

Statistical Modelling & Predictive Analytics – University of Calgary

2025

- Analysed a 2,598-participant dataset in R using exploratory analysis, logistic regression, interaction modelling, model selection, confidence intervals, and statistical diagnostics.
- Evaluated predictive performance using ROC/AUC and compared logistic regression with Random Forest and Decision Tree models, balancing predictive performance against interpretability.
- Applied longitudinal-data techniques including repeated-measures analysis, mixed-effects models, and GEE, with emphasis on model diagnostics, within-subject dependence, and communicating results.